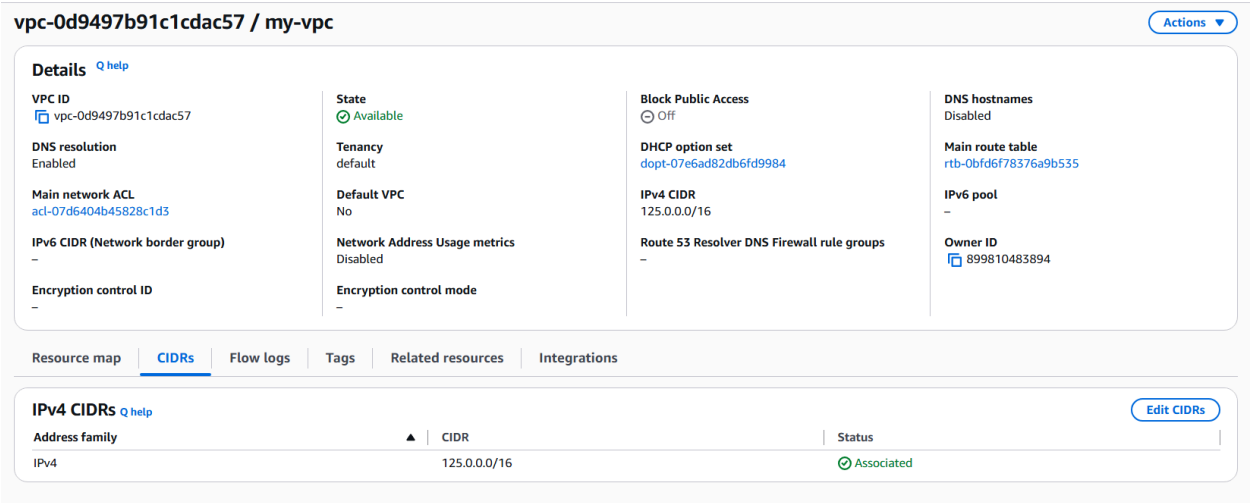
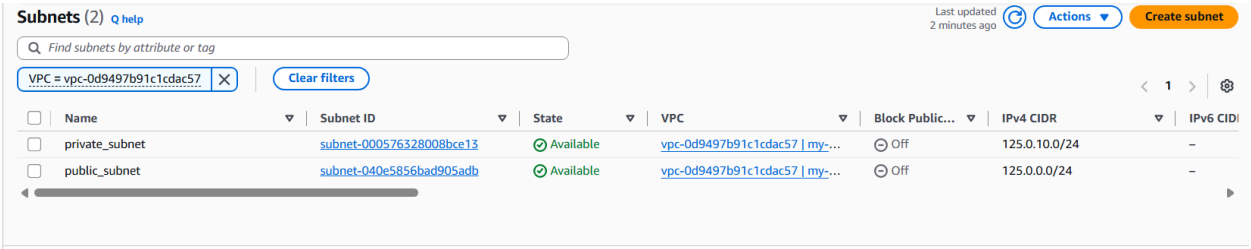
Configuring VPC in AWS

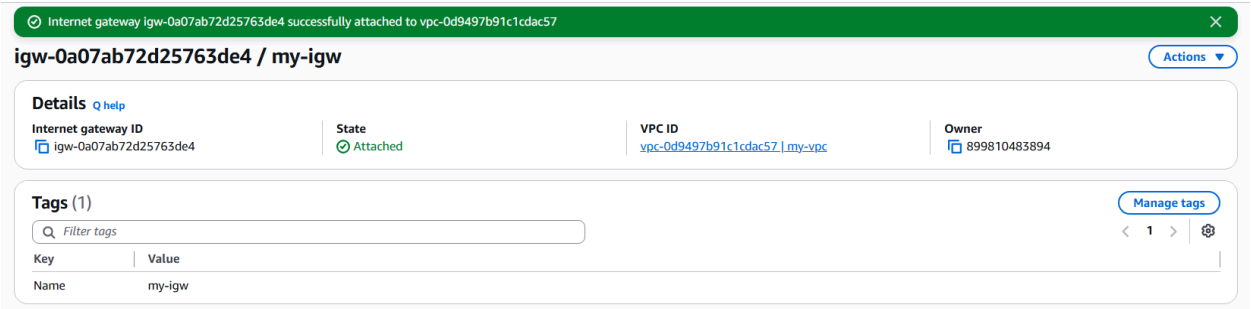
1. Creating Standard VPC:



2. Provision the VPC with 2 Subnets:

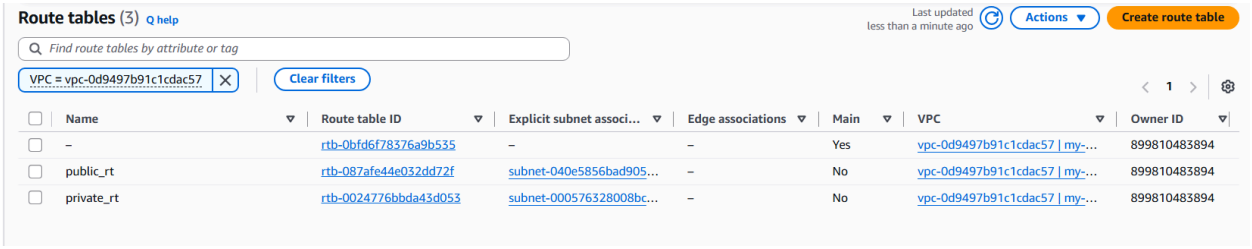


3. Creating Internet Gateway:



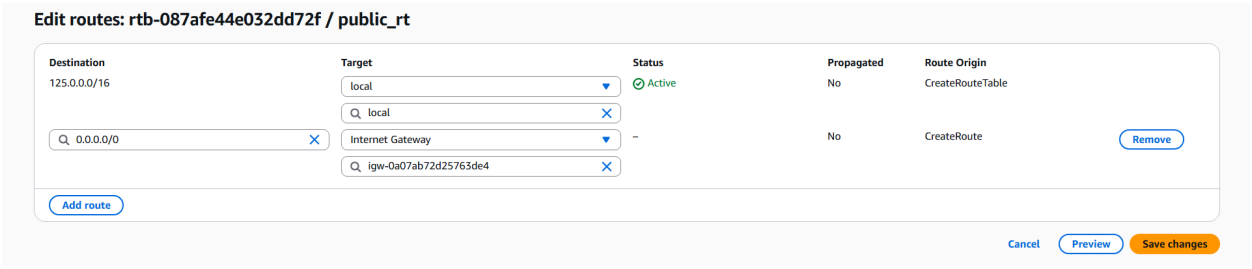
The internet gateway is manually assigned to the VPC created. This provides our network with connection to the internet.

4. Create Route Tables

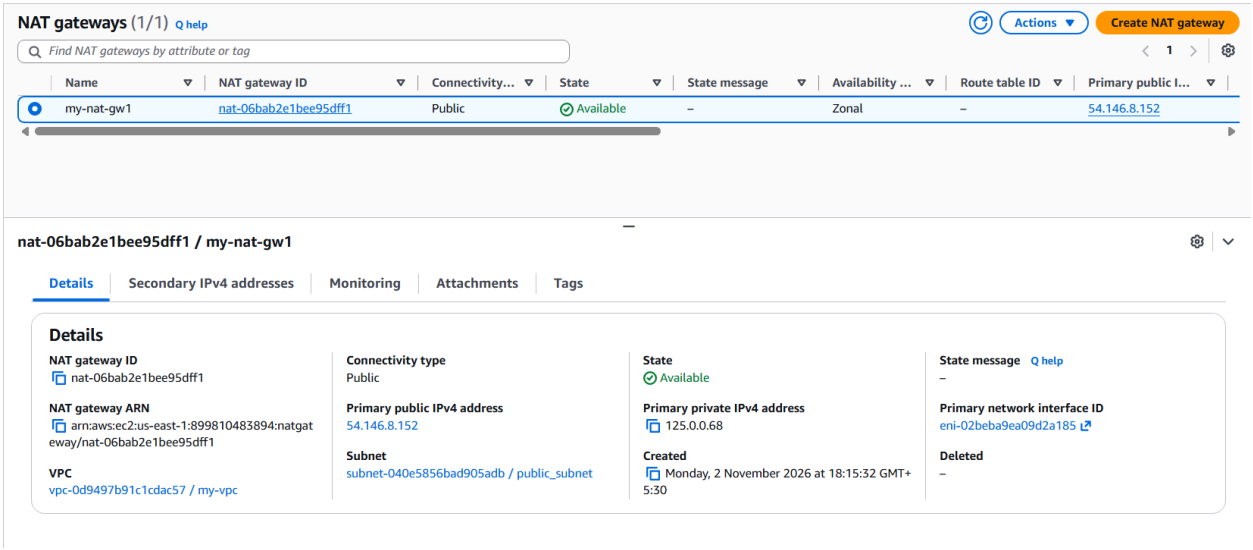


Two route tables are created, one for each subnet. We must configure the subnet association manually for each route table.

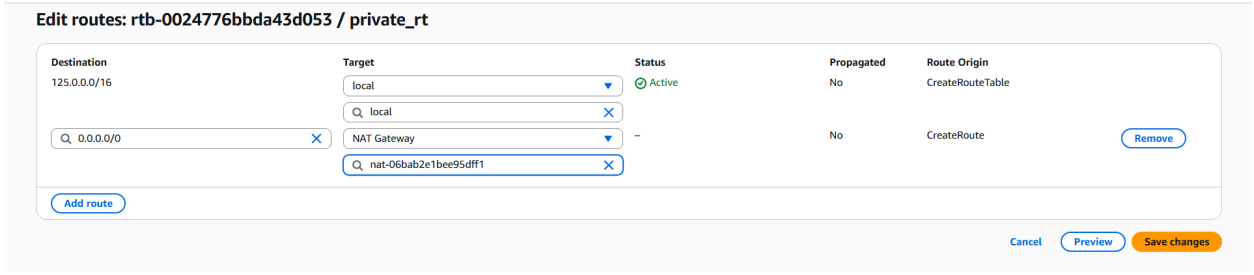
5. Adding IGW to Public RT Routes:



6. Create NAT Gateway:

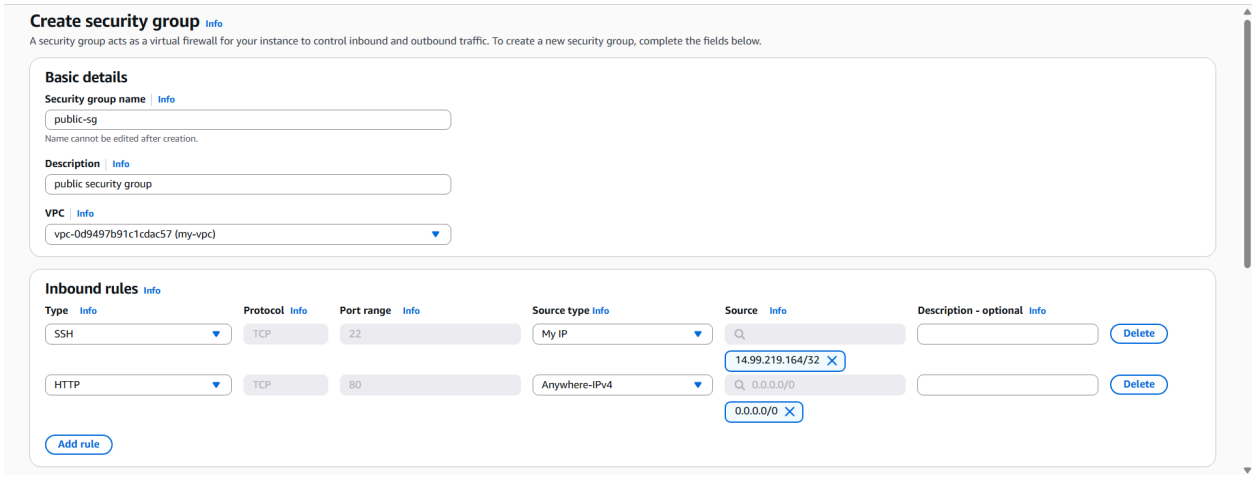


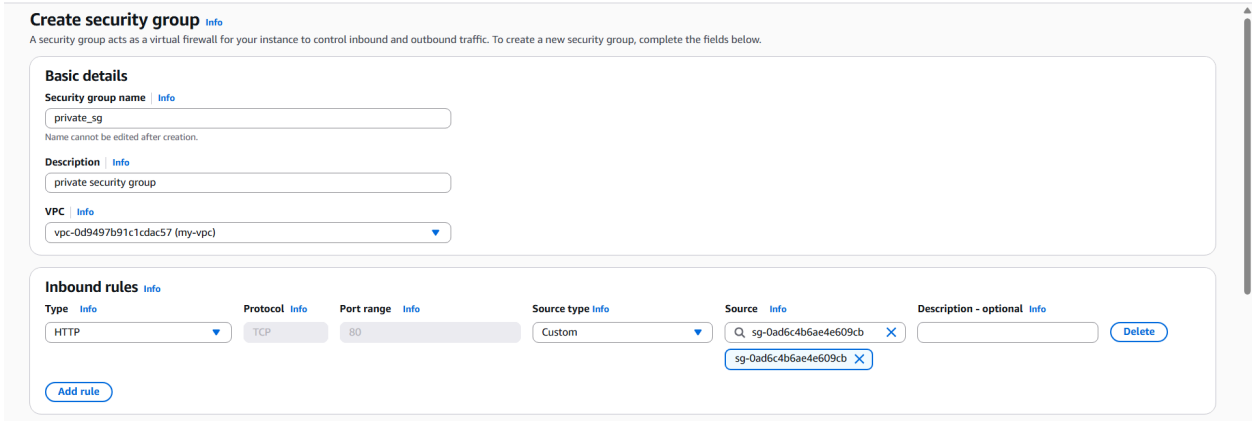
The private subnet must be connected to the NAT Gateway. We do this by adding a route from the private route table to the NAT Gateway.

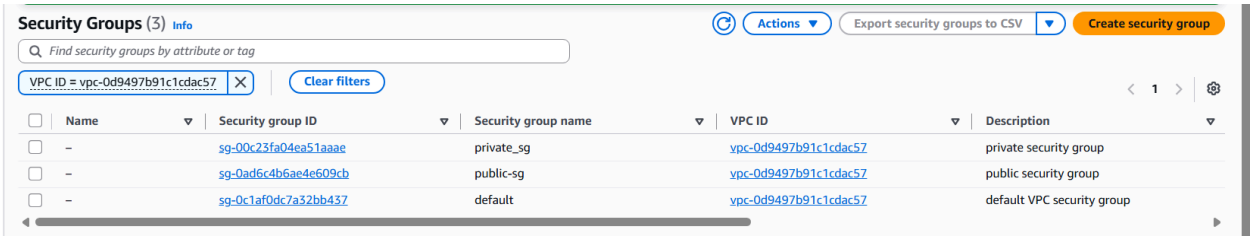


7. Creating Security Groups:

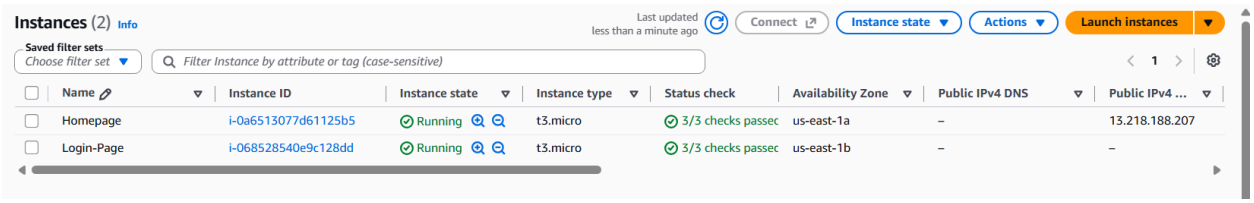
We create a security group each for the two subnets we created. Hence, one public security group and one private security group.







8. Creating EC2 Instances:



The Homepage server is configured to the public subnet we created. The Login-Page server is configured to the private subnet. Also we disable auto assignment of public IPv4 to the private EC2 server.

9. Installing Apache in both Servers:

```
[ec2-user@ip-125-0-0-30 ~]$ sudo systemctl enable --now httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-125-0-0-30 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-11-02 13:10:41 UTC; 6s ago
       Docs: man:httpd.service(8)
   Main PID: 10966 (httpd)
     Status: "Started, listening on: port 80"
      Tasks: 177 (limit: 1059)
     Memory: 13.9M
        CPU: 78ms
     CGroup: /system.slice/httpd.service
             ├─10966 /usr/sbin/httpd -DFOREGROUND
             ├─11106 /usr/sbin/httpd -DFOREGROUND
             ├─11110 /usr/sbin/httpd -DFOREGROUND
             ├─11111 /usr/sbin/httpd -DFOREGROUND
             └─11112 /usr/sbin/httpd -DFOREGROUND

Nov 02 13:10:41 ip-125-0-0-30.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Nov 02 13:10:41 ip-125-0-0-30.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Nov 02 13:10:41 ip-125-0-0-30.ec2.internal httpd[10966]: Server configured, listening on: port 80
[ec2-user@ip-125-0-0-30 ~]$
```

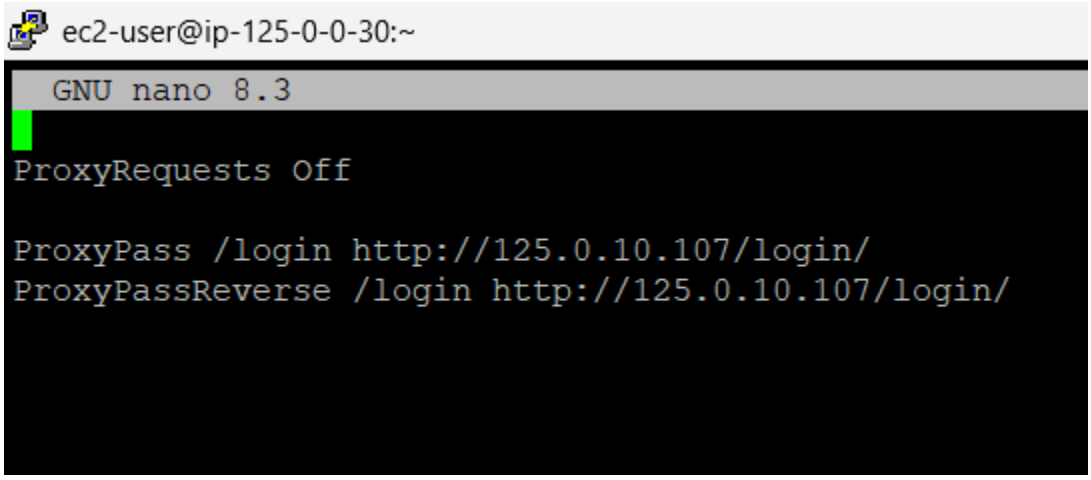
```
[ec2-user@ip-125-0-10-107 ~]$ sudo systemctl enable --now httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-125-0-10-107 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-11-02 15:43:23 UTC; 10s ago
       Docs: man:httpd.service(8)
   Main PID: 17952 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1059)
     Memory: 14.1M
        CPU: 83ms
     CGroup: /system.slice/httpd.service
             ├─17952 /usr/sbin/httpd -DFOREGROUND
             ├─18102 /usr/sbin/httpd -DFOREGROUND
             ├─18108 /usr/sbin/httpd -DFOREGROUND
             ├─18153 /usr/sbin/httpd -DFOREGROUND
             └─18176 /usr/sbin/httpd -DFOREGROUND

Nov 02 15:43:23 ip-125-0-10-107.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Nov 02 15:43:23 ip-125-0-10-107.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Nov 02 15:43:23 ip-125-0-10-107.ec2.internal httpd[17952]: Server configured, listening on: port 80
[ec2-user@ip-125-0-10-107 ~]$
```

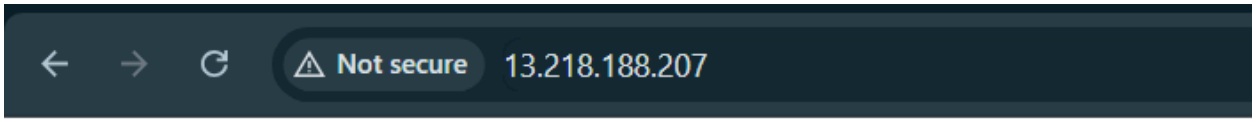
10. Setting up Apache Proxy

We create separate html files in each of the servers. The public server gets a homepage layout, which has a button, on clicking should redirect to a login page. The login page, in turn, is stored in the private server.

For this, we set up a Reverse Proxy



11. Testing Connection Between Public and Private EC2:

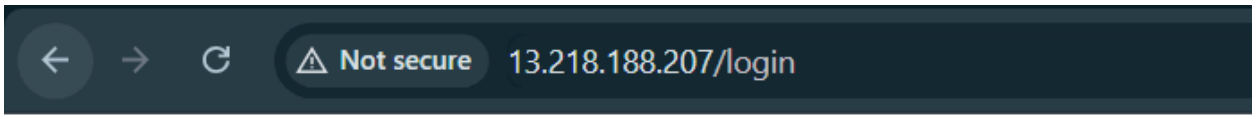


Welcome to the Homepage Server

This page is hosted on the Public EC2 instance.

Login

The image above is the webpage accessed through the public server's public IPv4 address. On clicking the "Login" button, the reverse proxy we set up must deliver the login page from the private server.



Welcome to the Private Login Server

This page is hosted on the Private EC2 instance.

Username:

Password:

Login

As seen here, the private login server page is reached, without ever exposing the private IPv4 address of the private EC2 server.